

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THE UNIVERSITY OF TEXAS AT ARLINGTON

PROJECT CHARTER

CSE 4316 / CSE 4317: SENIOR DESIGN

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AgriHome

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REVISION HISTORY

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0.3	02.18.2026	Team	Release candidate
1.0	02.18.2026	Team	Official release
1.1	02.18.2026	Team	Customer-requested revisions
2.0	07.26.2026	Team	Comprehensive project charter revision

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1 PROBLEM STATEMENT

Plant health management remains a significant challenge for farmers, gardeners, greenhouse operators, and educational research teams. Plant disease, pest stress, nutrient imbalance, and environmental problems often begin with subtle symptoms such as spotting, discoloration, wilting, or leaf-shape changes. Early symptoms may be overlooked or mistaken for nutrient deficiencies or environmental stress, especially when plants are inspected manually across several trays or shelves.

Accurate plant-health assessment requires specialized knowledge and regular observation. As a result, many growers are unable to identify plant-health issues early or accurately. Delayed recognition can allow infection to spread, reduce yield, damage experiments, and increase unnecessary chemical treatment. The need is not only for a disease classifier, but for a repeatable monitoring workflow that can capture clear images of individual plants over time and present those records to an operator.

AgriHome addresses this need by combining a rack-mounted camera system, edge-device communication, durable image records, and a web-based Vision Console. Providing timely plant-health indications allows users to respond more effectively, compare plant images over time, and make better decisions about care, isolation, watering, lighting, and follow-up inspection.

2 METHODOLOGY

AgriHome is a rack-based plant monitoring system that captures individual plant images through a camera mounted on an XY motion mechanism. The camera is attached to the plant shelf structure and moves horizontally along the X-axis and vertically along the Y-axis to predefined capture poses. At each pose, the camera faces one selected plant and captures a dedicated image rather than relying on a single wide view of the entire rack.

A Raspberry Pi edge device supports camera movement, image capture, device communication, and image transfer. Operators use the AgriHome Vision Console to manage plants and trays, monitor connected devices, initiate capture sequences, review captured images, and view available plant-health information. The application services maintain plant records, image metadata, device status, and capture history so that plant observations can be tracked over time.

When computer-vision processing is configured, captured images may be analyzed to surface plant-health indications, possible symptoms, or confidence-scored results for operator review. The charter treats this capability as decision support rather than a guaranteed diagnosis. The completed workflow therefore emphasizes repeatable capture, reliable device status, accessible records, and clear operator review before any automated analysis is treated as final.

3 VALUE PROPOSITION

AgriHome offers value by making plant monitoring more repeatable, accessible, and useful for everyday operators. Greenhouse operators and growers can reduce manual inspection effort by capturing consistent images of individual plants and reviewing them through a single console. Home gardeners and small-scale growers receive a practical tool for monitoring plant changes without needing advanced agricultural equipment.

For researchers and educational institutions, the project strengthens interdisciplinary learning and applied research by integrating software engineering, embedded systems, motion control, data management, and computer vision. The rack-mounted XY capture system creates a structured path for collecting plant images under repeatable positions, which can support future studies of plant growth, disease progression, and image-based analysis.

For future product stakeholders, AgriHome demonstrates a scalable direction for plant monitoring: edge hardware captures plant-level data, web services preserve records, and operators interact with the system through a clear Vision

Console. The project supports sustainable agriculture goals by encouraging earlier response, better documentation, and more informed plant-care decisions.

4 DEVELOPMENT MILESTONES

The following milestones summarize the project progression across the Spring/Summer 2026 effort. The table uses verified month-level dates where available and team-review month targets for remaining deliverables.

Milestone	Target or Verified Date
Project Charter first draft	February 2026
System Requirements Specification	March 2026
Architectural Design Specification	April 2026
Initial plant-health detection and test-rig demonstration	April 2026
Vision Console baseline with authentication, dashboard, trays, plants, and persistent records	June 2026
Raspberry Pi edge-device integration with registration, heartbeat, and device status	June 2026
Demonstration of rack-mounted XY camera movement	July 2026
Demonstration of individual plant image capture at predefined poses	July 2026
Demonstration of Vision Console device management and capture workflow	July 2026
Demonstration of integrated image transfer, storage, and review	July 2026
Detailed Design Specification	July 2026
College of Engineering Innovation Day poster presentation	July 2026
Final project demonstration and closeout submission	July 2026

Table 1: Development milestones

5 BACKGROUND

Plant health monitoring is difficult because meaningful changes can be small, gradual, and unevenly distributed across plants. Manual inspection works for very small collections but becomes inefficient as trays, shelves, and growth cycles increase. Fixed cameras can capture wide contextual views, yet they may not provide consistent plant-level images when leaves overlap, shelves cast shadows, or plants are positioned at different distances from the camera.

AgriHome expands the monitoring problem from image classification alone to an end-to-end system-integration challenge. The project includes a plant rack, a camera carriage attached to the shelf structure, XY motion to repeatable capture poses, Raspberry Pi edge communication, application services, image ingest, storage, and an operator-facing Vision Console. This combination allows the system to capture one plant at a time, associate each image with a known plant record, and review historical changes without depending entirely on manual photo-taking.

Computer vision remains an important supporting component. Image-based plant disease research has shown that convolutional neural networks and curated plant-image datasets can assist with symptom recognition under controlled conditions. AgriHome uses that research direction carefully by treating model output as plant-health information that may support operator decisions, while the prototype's core contribution is the repeatable capture and monitoring workflow.

The senior-design project is sponsored by Professor Noor. The intended users include home gardeners, greenhouse operators, student researchers, and educational agricultural programs that need a practical and understandable plant-monitoring prototype. The project aligns with applied computing, sustainable technology, and interdisciplinary

engineering goals.

6 RELATED WORK

Image-based plant disease detection has been widely studied. Mohanty, Hughes, and Salathe demonstrated that deep learning could classify plant diseases from images using the PlantVillage dataset under controlled data conditions [1]. Ferentinos developed convolutional neural-network models for plant disease detection and diagnosis using leaf images across many crop-disease combinations [2]. Hughes and Salathe released a large open repository of plant-health images intended to support mobile disease diagnostics and machine-learning research [3].

Related work in greenhouse monitoring and smart agriculture supports AgriHome's system-level direction. IoT approaches for smart greenhouses emphasize sensing, monitoring, control, and data exchange across distributed devices [4]. Reviews of smart greenhouses in the era of IoT and AI describe the value of combining digital monitoring, sensing, and intelligent services for controlled-environment agriculture [5]. These areas support AgriHome's use of connected edge hardware and a web console rather than a stand-alone image classifier.

Plant phenotyping research is also relevant because it emphasizes repeatable imaging and the value of structured image collection. Cho and Yang describe a low-cost camera-network approach for plant-factory phenotyping [6], while plant-phenotyping reviews discuss automated imaging as a way to measure plant traits more consistently than manual inspection [7]. AgriHome adapts this repeatable-imaging idea to a smaller educational prototype with a rack-mounted XY camera and individual plant capture.

7 SYSTEM OVERVIEW

AgriHome consists of a physical capture system, an edge-device communication layer, application services, persistent records, and an operator-facing Vision Console. Plants are placed on a multi-level rack or shelf. The camera is attached to the shelf structure and moves along X-axis and Y-axis paths so it can stop at predefined capture poses and face one plant at a time.

At a high level, the operating workflow is:

- The operator uses the Vision Console to manage trays, plants, connected devices, and capture actions.
- The Raspberry Pi edge device reports heartbeat and status information to the application services.
- The operator selects a plant, tray, or capture sequence from the Vision Console.
- The camera moves along the X-axis and Y-axis to a predefined capture pose.
- The camera stops and captures an image of the selected individual plant.
- The edge device transfers the image and metadata to the AgriHome application.
- The application stores the image record, plant association, timestamp, and device information.
- The Vision Console displays the image, history, device status, and related plant information.
- Optional computer-vision services analyze the image and return plant-health results for review.

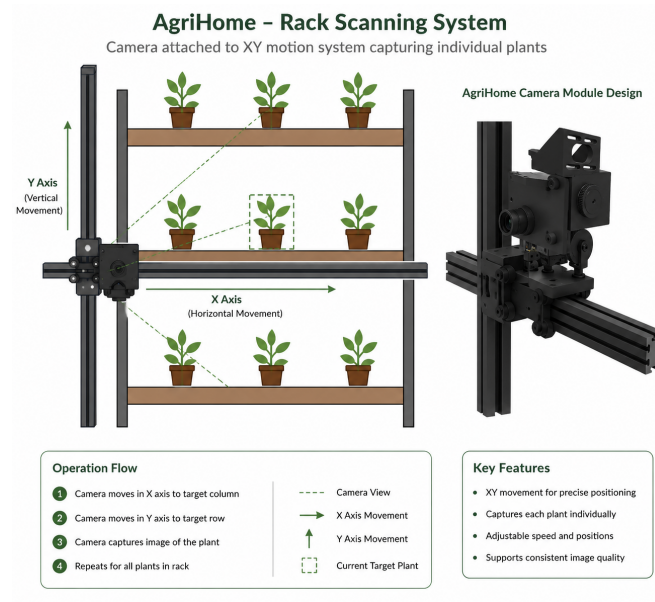


Figure 1: Hardware Setup

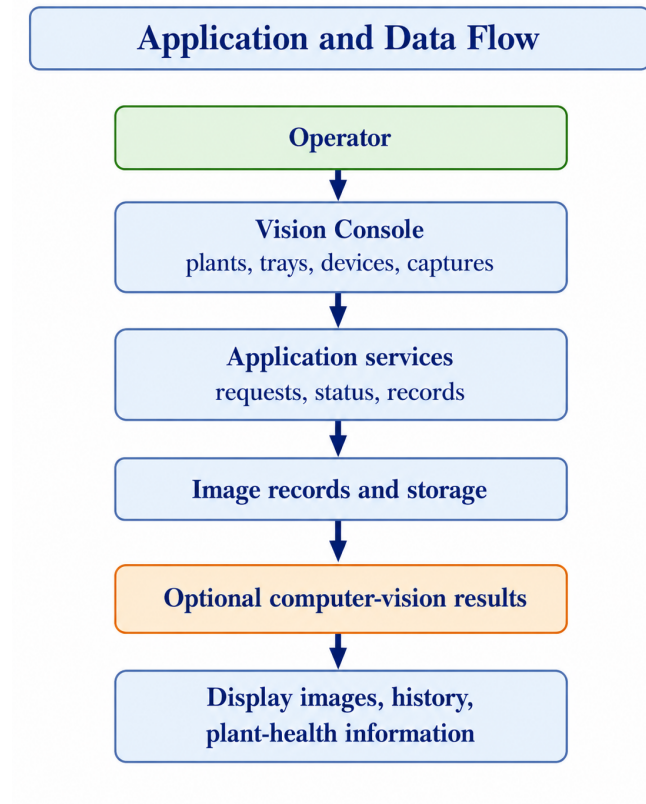


Figure 2: System Overview

8 ROLES & RESPONSIBILITIES

The following verified assignments describe the major project responsibilities. Responsibilities may overlap when integration, testing, or demonstrations require collaboration across subsystems.

HARDWARE AND MOTION SYSTEM

Assigned members: Thuc Dang; Muhammad Saad Memon.

Responsibilities include plant-rack setup, rack-mounted camera carriage, XY movement, hardware assembly, travel limits, actuator testing, wiring, and physical integration.

FRONTEND AND VISION CONSOLE

Assigned members: Sagar Khadka; Laxman Bhushal.

Responsibilities include the dashboard, plants and trays interfaces, device-management views, capture controls, image history, responsive design, and operator workflows.

BACKEND AND DATABASE

Assigned member: Laxman Bhushal.

Responsibilities include application services, API behavior, authenticated routes, plant and capture records, device records, image metadata, database integration, and storage references.

EDGE-DEVICE INTEGRATION

Assigned members: Muhammad Saad Memon; Nathan Nguyen Tran.

Responsibilities include Raspberry Pi setup, device registration, heartbeat reporting, movement commands, image capture, image transfer, status reporting, and recovery procedures.

COMPUTER VISION

Assigned members: Govinda Awasthi; Laxman Bhushal.

Responsibilities include image-processing research, model integration, model evaluation, confidence reporting, plant-health result presentation, and documentation of model limitations.

SCRUM COORDINATION AND DOCUMENTATION

Assigned members: Alex Tablizo; Sagar Khadka.

Responsibilities include sprint coordination, Jira backlog maintenance, sprint planning support, sponsor communication, meeting notes, milestone tracking, documentation updates, and deliverable review.

9 COST PROPOSAL

The project budget allocated by the customer is \$800 through the CSE Department of The University of Texas at Arlington. The budget supports the prototype hardware and materials required to construct, test, and demonstrate the AgriHome rack-scanning system. The current estimated cost remains within the approved allocation.

9.1 PRELIMINARY BUDGET

Item	Estimated Cost	Notes
Camera / webcam module	\$70	
Computer (local / Pi-class edge)	\$190	
Computer (server)	\$0	Used available teammate's server
Camera fixture	\$50	
Plant fixture / tray materials	\$100	
Plants	\$20	
Misc. networking / mounting	\$50	
Other 3D prints	\$0	Provided by UTA
Estimated Total	\$480	Within the \$800 project allocation

Table 2: Preliminary budget

9.2 CURRENT & PENDING SUPPORT

The primary funding source is the CSE Department of The University of Texas at Arlington. Additional funding may be requested through the project sponsor or department if approved and necessary. No informal commitment is treated as guaranteed funding in this charter.

10 FACILITIES & EQUIPMENT

The project uses university resources and team development equipment for implementation, testing, and demonstration. The Senior Design lab provides meeting space, computer access, stable power, and a controlled indoor environment for application testing and prototype review. Makerspace resources may support 3D printing, bracket fabrication, rack modifications, and assembly of the camera carriage when available and approved.

AgriHome also requires safe indoor storage for the plant rack, repeatable lighting conditions for image capture, test plants, a stable electrical supply, and local network access for the Raspberry Pi edge device. Team computers support web application development, database testing, computer-vision experimentation, and documentation. The final setup should be arranged so that moving components remain visible, reachable for maintenance, and safe for demonstration.

11 ASSUMPTIONS

- The camera, Raspberry Pi edge device, and selected motion hardware remain available and functional during integration and demonstration.
- The rack-mounted camera carriage can be mechanically aligned so predefined capture poses remain repeatable over time.
- The XY motion range is sufficient to capture the intended plants without exceeding physical travel limits.
- Power and network connectivity are stable enough for device heartbeat, capture commands, and image transfer during demonstrations.
- Lighting conditions can be controlled or documented to reduce extreme image-quality variability.
- The application has sufficient storage for captured images and related metadata.

- Operators have authorized access to the Vision Console and understand that automated plant-health outputs are decision support.
- Sufficient plants and testing time are available to verify rack setup, capture poses, image history, and device status behavior.
- Publicly available or team-collected images may be used for optional computer-vision evaluation according to course and university policy.

12 CONSTRAINTS

The following constraints define practical boundaries for the Spring/Summer 2026 prototype and documentation effort.

- The final prototype and required documentation must be completed by the applicable Spring/Summer 2026 course deadline.
- Total development costs must remain within the allocated \$800 budget unless additional sponsor or department funding is formally approved.
- Development follows the sprint cadence, review schedule, and submission requirements defined by the senior-design course.
- The prototype must operate within available university lab access, equipment availability, storage rules, and safety policies.
- The rack dimensions, rail length, carriage design, and camera focus distance limit the possible capture positions.
- The moving camera mechanism must operate safely around plants, wiring, shelves, and users.
- Consumer-grade cameras, Raspberry Pi-class edge hardware, and practical network connectivity limit prototype reliability.
- Only approved public datasets or self-collected images may be used for optional model training and evaluation.
- Team size and development hours are fixed by course enrollment and academic commitments.

13 RISKS

Risk exposure is calculated as $Exposure = Probability \times Loss$. The table preserves the original UTA Project Charter risk-table structure. Asterisks indicate preliminary placeholder values for team review where verified estimates were unavailable.

Risk Description	Probability	Loss (days)	Exposure (days)
XY motion mechanism misalignment	0.40*	12	4.80
Application-to-device integration defect	0.35*	10	3.50
Stepper motor, driver, or power failure	0.25*	12	3.00
Camera capture failure or poor image quality	0.30*	8	2.40
Raspberry Pi network disconnection	0.30*	7	2.10
Image upload interruption	0.25*	7	1.75
Schedule conflicts	0.20*	7	1.40

Table 3: Overview of highest-exposure project risks

Mitigation strategies include early mechanical alignment checks, low-speed motion testing, spare-part planning where possible, repeatable lighting checks, visible device heartbeat status, capture-status tracking, upload retry procedures, integration testing before demonstrations, and consistent Jira sprint tracking.

14 DOCUMENTATION & REPORTING

Documentation and reporting materials are maintained as controlled project artifacts. They capture project scope, requirements, design decisions, implementation details, sprint progress, testing evidence, and closeout deliverables.

14.1 MAJOR DOCUMENTATION DELIVERABLES

Major documentation deliverables are reviewed at sprint or milestone checkpoints and updated when requirements, architecture, design details, risk, or implementation status changes.

14.1.1 PROJECT CHARTER

The Project Charter defines project vision, scope, stakeholders, responsibilities, cost assumptions, facilities, constraints, risks, documentation practices, and closeout expectations. It is version-controlled through the revision history and reviewed when scope, schedule, system design, or sponsor expectations change. Initial version delivery: Sprint 1. Final version delivery: prior to final project submission.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification defines functional requirements, nonfunctional requirements, interfaces, user workflows, device interactions, and verification expectations. It is updated when requirements are approved, clarified, or changed through sprint reviews, testing, or stakeholder feedback. The final SRS should distinguish operator workflows, edge-device behavior, data storage, motion/capture requirements, and optional computer-vision behavior.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification describes the high-level system structure, major components, interfaces, data flow, deployment context, and integration boundaries. For AgriHome, the ADS should represent the rack-mounted XY capture hardware, Raspberry Pi edge device, Vision Console, application services, database/storage layer, and optional computer-vision service.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification describes internal module behavior, data structures, workflows, algorithms, component responsibilities, and implementation-level decisions. It should reflect the actual Vision Console, backend services, database records, Raspberry Pi edge workflow, motion/capture behavior, image ingest, and optional CV integration rather than obsolete schema or fixed-camera assumptions.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

The team maintains the product backlog and sprint tasks in Jira. Backlog items are refined continuously, prioritized by stakeholder value, technical risk, and dependencies, and assigned to team members during sprint planning.

14.2.2 SPRINT PLANNING

During Sprint Planning, the team reviews priority work, estimates capacity, identifies dependencies, and selects backlog items that can realistically be completed. Planning considers hardware bench time, motion-system availability, network access, device integration, Vision Console work, database changes, testing, and documentation.

14.2.3 SPRINT GOAL

The sprint goal is developed collaboratively during Sprint Planning. The Scrum Master facilitates the discussion, while the team considers stakeholder feedback, project priorities, dependencies, and available capacity. A good sprint goal describes a testable outcome, such as proving that the edge device reports heartbeat status or that the camera captures and stores an individual plant image.

14.2.4 SPRINT BACKLOG

The sprint backlog contains the Jira tasks committed for the sprint. Each task has an owner, acceptance criteria, status, and relationship to the sprint goal. The backlog is updated as work progresses, blockers appear, or tasks are completed.

14.2.5 TASK BREAKDOWN

Stories are decomposed into implementable tasks such as hardware assembly, motion testing, camera capture verification, device heartbeat, API work, database updates, Vision Console views, image-storage validation, CV experiments, deployment notes, and documentation updates.

14.2.6 SPRINT BURNDOWN CHARTS

Sprint burndown charts visualize remaining work versus time. The team may use Jira burndown reports or an equivalent exported chart during sprint coordination and reviews. The example below shows the expected format only; it does not claim to represent actual team data.

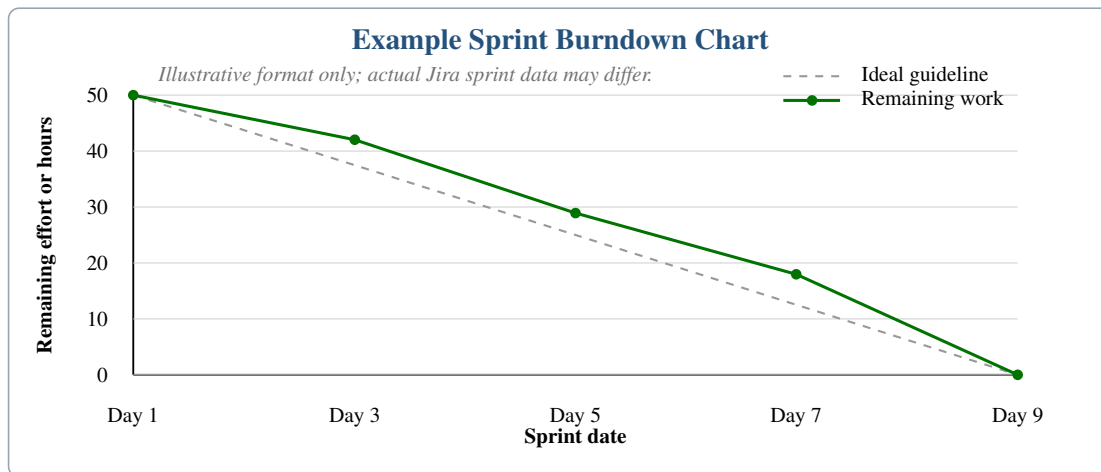


Figure 3: Format for a sprint burndown chart.

14.2.7 SPRINT RETROSPECTIVE

Sprint retrospectives review what worked, what did not work, and what should change in the next sprint. Action items may include earlier hardware testing, clearer acceptance criteria, improved Jira updates, better integration checkpoints, or more complete documentation.

14.2.8 INDIVIDUAL STATUS REPORTS

Individual status reports document each member's completed work, current work, blockers, and planned next steps. The report will be submitted after the current sprint and before the next sprint begins.

14.2.9 ENGINEERING NOTEBOOKS

Each team member will update an engineering notebook at least once per sprint and after significant design, implementation, or testing activity. Entries will document objectives, decisions, diagrams, hardware changes, software changes, test results, issues, and next steps. Another team member working on the same subsystem will review or witness entries when required, and notebook evidence will be referenced during sprint reviews and closeout preparation.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

The system prototype includes the plant rack, rack-mounted XY motion system, camera carriage, Raspberry Pi edge device, Vision Console, backend application services, storage/database layer, and optional computer-vision service. Any formal prototype acceptance procedure will be determined with the project sponsor before closeout.

14.3.2 PROJECT POSTER

The project poster will summarize the problem, system design, major hardware and software components, key features, implementation results, and team information. Final dimensions and delivery requirements will follow the College of Engineering Innovation Day guidelines.

14.3.3 WEB PAGE

A project webpage will be provided at closeout. It will include a project overview, team information, system features, selected diagrams, demonstration media, and approved project documentation. Public accessibility will depend on university and repository policies.

14.3.4 DEMO VIDEO

The demonstration video will present the AgriHome system overview, camera movement, individual plant capture, device monitoring, Vision Console operation, image history, and available plant-health analysis. The final video length and delivery format will follow course requirements.

14.3.5 SOURCE CODE

Source code will be maintained in the team's Git repository using Git-based version control. At closeout, the repository will include the web application, backend services, database configuration, edge-device software, optional computer-vision components, deployment instructions, and supporting documentation, subject to university and project access policies.

14.3.6 SOURCE CODE DOCUMENTATION

Source code documentation will use a mixed strategy appropriate to the implementation. Documentation may include README files, API documentation, inline TypeScript documentation, typed interfaces, Python docstrings, deployment notes, database schema documentation, configuration examples, and generated documentation where appropriate.

14.3.7 HARDWARE SCHEMATICS

Wiring diagrams will document power connections, motor drivers, camera connections, Raspberry Pi interfaces, sensors if used, and applicable communication protocols. Hardware notes should also describe rack mounting, cable routing, travel limits, and safe operating practices for the moving camera carriage.

14.3.8 CAD FILES

STEP files and STL files for applicable 3D-printed or fabricated components will be included in the project repository when available. CAD materials should identify camera-carriage parts, brackets, mounting fixtures, rail supports, and any rack-specific hardware designed by the team.

14.3.9 INSTALLATION SCRIPTS

Deployment scripts and setup instructions will be provided for the web application, database, Raspberry Pi edge software, required configuration, image-storage directories, and optional computer-vision services. The documentation will distinguish between development, demonstration, and deployment environments.

14.3.10 USER MANUAL

The user manual will explain authentication, Vision Console navigation, plant and tray management, device registration, device status and heartbeat, actuator limits, capture pose sequences, camera movement, Take Picture operation, image history, offline-device troubleshooting, rack and camera setup, safe motion operation, and basic system recovery.

15 REFERENCES

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